# Dr. Dhruv Kumar

**BIO-DATA**

Assistant Professor, BITS Pilani

dhruv.kumar@pilani.bits-pilani.ac.in
(+91) 77427 10554

## Personal Details

| Field | Details |
|---|---|
| **Name** | Dr. Dhruv Kumar |
| **Address** | 6120-F, New Academic Building, BITS Pilani, Rajasthan – 333031 |
| **Institution** | Birla Institute of Technology & Science (BITS), Pilani |
| **Email** | dhruv.kumar@pilani.bits-pilani.ac.in |
| **Phone** | (+91) 77427 10554 |
| **Date of Birth** | 11 January 1994 |
| **Gender** | Male |
| **Category** | General |
| **Differently Abled** | No |
| **ORCID** | 0000-0003-4831-1847 |

## Academic Qualifications

| # | Degree | Year | Subject | University / Institution | CGPA |
|---|---|---|---|---|---|
| 1 | B.E. (Hons.) | 2014 | Computer Science | Birla Institute of Technology and Science, Pilani | 9.92/10 |
| 2 | Ph.D. | 2022 | Computer Science & Engineering | University of Minnesota, Twin Cities, USA | 4.0/4.0 |

## Work Experience

| # | Position | Institution | From | To |
|---|---|---|---|---|
| 1 | Assistant Professor | BITS Pilani | May 2024 | Present |
| 2 | Adjunct Faculty | IIIT Delhi | May 2024 | Present |
| 3 | Assistant Professor | IIIT Delhi | Aug 2022 | May 2024 |
| 4 | Post-Doctoral Researcher | Microsoft Research, Bengaluru | Mar 2022 | Jul 2022 |
| 5 | Ph.D. Intern | Google Cloud, California, USA | May 2019 | Aug 2019 |
| 6 | Software Engineer / Quant Analyst | Goldman Sachs, Bengaluru | Nov 2014 | Apr 2016 |
| 7 | Summer Intern | CSIR-CEERI, Pilani | May 2012 | Jul 2012 |

## Professional Recognition / Awards / Prizes / Fellowships

| # | Award / Recognition | Awarding Body | Year |
|---|---|---|---|
| 1 | Teaching Excellence Award (AY 2023–24) | IIIT Delhi | Sep 2024 |
| 2 | Teaching Excellence Award (AY 2022–23) | IIIT Delhi | Sep 2023 |
| 3 | 3M Science and Technology Fellowship | University of Minnesota, Twin Cities | 2017–2021 |
| 4 | Travel Grant | USENIX ATC | 2019 |
| 5 | Travel Grant | ACM SIGMETRICS | 2019 |
| 6 | Travel Grant | USENIX OSDI | 2018 |
| 7 | Merit Scholarship (Top 10 among 800 students) | BITS Pilani | 2010–2014 |
| 8 | Outstanding Undergraduate Thesis | BITS Pilani | 2014 |

## Publications (in Year-wise Descending Order)

*Author's name in* ***bold****.* ***[Best Paper Award]*** *denotes prize-winning papers.*

### Journal Articles & Transactions

**J1.** Alakh Sharma, Gaurish Trivedi, Kartikey Singh Bhandari, Yash Sinha, **Dhruv Kumar**, Pratik Narang, and Jagat Sesh Challa. 2026. Generative Evolutionary Meta-Solver (GEMS): Scalable Surrogate-Free Multi-Agent Reinforcement Learning. **Transactions on Machine Learning Research (TMLR) 2026**.

**J2.** Raj Gupta, Harshita Goyal, **Dhruv Kumar**, Apurv Mehra, Sanchit Sharma, Kashish Mittal, and Jagat Sesh Challa. 2026. Sakshm AI: Advancing AI-Assisted Coding Education for Engineering Students in India Through Socratic Tutoring and Comprehensive Feedback. **ACM Journal on Computing and Sustainable Societies (JCSS) 2026**.

**J3.** Harshita Goyal, Garima Garg, Prisha Mordia, Veena Ramachandran, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. The Impact of Large Language Models on K-12 Education in Rural India: A Thematic Analysis of Student Volunteers' Perspectives. **Scientific Reports (2025)**.

**J4.** Balamurugan Palanisamy, Vikas Hassija, Arpita Chatterjee, Arpita Mandal, Debanshi Chakraborty, Amit Pandey, G. Sai Sesha Chalapathi, and **Dhruv Kumar**. 2025. Transformers for Vision: A Survey on Innovative Methods for Computer Vision. **IEEE Access 13 (2025)**, 95496–95523.

**J5.** Harshal Akolekar, Piyush Jhamnani, Vikash Kumar, Vinay Tailor, Aditya Pote, Ankit Meena, Kamal Kumar, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. The Role of Generative AI Tools in Shaping Mechanical Engineering Education from an Undergraduate Perspective. **Scientific Reports 15 (2025)**, Article 9214.

**J6.** Ashutosh Sharma, **Dhruv Kumar**, Srinivas Rallapalli, et al. 2023. Wetland Functional Assessment and Uncertainty Analysis Using Fuzzy Alpha-Cut–Based Modified Hydrogeomorphic Approach. **Environmental Science & Pollution Research 30 (2023)**, 72900–72915.

**J7.** Ananya Jain, Srinivas Rallapalli, and **Dhruv Kumar**. 2022. Cloud-Based Neuro-Fuzzy Hydro-Climatic Model for Water Quality Assessment under Uncertainty and Sensitivity. **Environmental Science & Pollution Research 29 (2022)**, 65259–65275.

**J8.** Poonam Goyal, Jagat Sesh Challa, **Dhruv Kumar**, Anuvind Bhat, Sundar Balasubramaniam, and Navneet Goyal. 2020. Grid-R-Tree: A Data Structure for Efficient Neighborhood and Nearest Neighbor Queries in Data Mining. **Int. J. Data Sci. Anal. 10 (2020)**, 25–47.

**J9.** **Dhruv Kumar**, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics. **Proc. ACM MACS 3, 2 (2019)**, 29:1–29:27.

**J10.** **Dhruv Kumar**, Poonam Goyal, and Navneet Goyal. 2018. An Efficient Method for Batch Updates in OPTICS Cluster Ordering. **Int. J. Data Anal. Tech. Strateg. 10 (2018)**, 57–80.

### Conference Papers

**C1.** Tulika Tewari, Nalin A. G. Arachchilage, Jagat Sesh Challa, and **Dhruv Kumar**. 2026. From Trust to Compromise: Outcome-Verified LLM Phishing Simulation and Real-Time Defense. In **Association for Computational Linguistics (ACL 2026)**.

**C2.** Vikranth Udandarao, Noel Abraham Tiju, Muthuraj Vairamuthu, Harsh Mistry, and **Dhruv Kumar**. 2026. Roamify: Designing and Evaluating an LLM-Based Google Chrome Extension for Personalised Itinerary Planning. In **The**

**Web Conference (WWW) 2026**.

**C3.** Pankaj Iiitd, **Dhruv Kumar**, Vinayak Abrol, and Vikram Goyal. 2026. BLEND: Balanced and Leaf-Enhanced Dual Fine-Tuning for Taxonomy Completion. In **The Web Conference (WWW) 2026**.

**C4.** Debdeep Sanyal, Agniva Maiti, Umakanta Maharana, **Dhruv Kumar**, Ankur Mali, C. Lee Giles, and Murari Mandal. 2025. Investigating Pedagogical Teacher and Student LLM Agents: Genetic Adaptation Meets Retrieval-Augmented Generation Across Learning Styles. In **EMNLP 2025**, 13359–13400.

**C5.** Varun Bharti, Shashwat Jha, **Dhruv Kumar**, and Pankaj Jalote. 2025. Combining Reasoning Optimised LLMs and SMT Solvers for Automated Loop Invariant Synthesis. In **AIWare 2025**. **[Best Paper Award]**

**C6.** Aditya Pathak, Rachit Gandhi, Vaibhav Uttam, Arnav Ramamoorthy, Pratyush Ghosh, Aaryan Raj Jindal, Shreyash Verma, Aditya Mittal, Aashna Ased, Chirag Khatri, Yashwanth Nakka, Devansh, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. Rubric Is All You Need: Improving LLM-Based Code Evaluation With Question-Specific Rubrics. In **ICER 2025**, 181–195.

**C7.** Adithya Chittem, Aishna Shrivastava, Sai Tarun Pendela, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. SAC: A Framework for Measuring and Inducing Personality Traits in LLMs with Dynamic Intensity Control. In **ICAART 2025**.

**C8.** Madhav Krishan Garg, Tejash Prasad, Tanmay Singhal, Chhavi Kirtani, Murari Mandal, and **Dhruv Kumar**. 2025. ReviewEval: An Evaluation Framework for AI-Generated Reviews. In **Findings of EMNLP 2025**, 20542–20564.

**C9.** Utkarsh Arora, Anupam Garg, Aryan Gupta, Samyak Jain, Ronit Mehta, Rupin Oberoi, Prachi, Aryaman Raina, Manav Saini, Sachin Sharma, Jaskaran Singh, Sarthak Tyagi, and **Dhruv Kumar**. 2025. Analyzing LLM Usage in an Advanced Computing Class in India. In **ACE '25**, 154–163.

**C10.** Ritvik Budhiraja, Ishika Joshi, Jagat Sesh Challa, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. "It's not like Jarvis, but it's pretty close!"—Examining ChatGPT's Usage among Undergraduate Students in CS. In **ACE 2024**, 124–133.

**C11.** Shubhang Shekhar Dvivedi, Vyshnav Vijay, Sai Leela Rahul Pujari, Shoumik Lodh, and **Dhruv Kumar**. 2024. A Comparative Analysis of Large Language Models for Code Documentation Generation. In **AIware 2024**.

**C12.** Harshal Dev, Raj Gupta, and **Dhruv Kumar**. 2024. From Cash to Cashless: UPI's Impact on Spending Behavior among Indian Users. In **CHI EA 2024**, 168:1–168:10.

**C13.** Shikhar Sharma, Manas Mhasakar, Apurv Mehra, Utkarsh Venaik, Ujjwal Singhal, **Dhruv Kumar**, and Kashish Mittal. 2024. Comuniqa: Exploring Large Language Models For Improving English Speaking Skills. In **COMPASS 2024**, 256–267.

**C14.** Ishika Joshi, Ritvik Budhiraja, Harshal Dev, Jahnvi Kadia, Mohammad Osama Ataullah, Sayan Mitra, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. ChatGPT in the Classroom: An Analysis of Its Strengths and Weaknesses for Solving Undergraduate CS Questions. In **SIGCSE 2024**, 625–631.

**C15.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin: Optimizing Geo-Distributed Joins using Aggregation-Based Transformations. In **CCGrid 2023**, 414–425. **[Best Paper Award]**

**C16.** Vikram Goyal, Naveen Kumar, Sourav S. Bhowmick, Pawan Goyal, Navneet Goyal, and **Dhruv Kumar** (Eds.). 2023. Big Data and Artificial Intelligence—11th International Conference, **BDA 2023**, Proceedings. Lecture Notes in Computer Science 14418. Springer.

**C17.** Joel Wolfrath, Nikhil Sreekumar, **Dhruv Kumar**, Yuanli Wang, and Abhishek Chandra. 2022. HACCS: Heterogeneity-Aware Clustered Client Selection for Accelerated Federated Learning. In **IPDPS 2022**, 985–995.

**C18.** Rallapalli Srinivas, Matt Drewitz, Joe Magner, Ajit Pratap Singh, **Dhruv Kumar**, and Yashwant Bhaskar Katpatal. 2022. Simulating Landscape Hydrologic Connectivity in a Precise Manner Using Hydro-Conditioning. In **Advances in Computational Modeling and Simulation**, Lecture Notes in Mechanical Engineering. Springer.

**C19.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2021. AggNet: Cost-Aware Aggregation Networks for Geo-Distributed Streaming Analytics. In **SEC 2021**, 297–311.

**C20.** Sonal Kumari, Poonam Goyal, Ankit Sood, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2017. Exact, Fast and Scalable Parallel DBSCAN for Commodity Platforms. In **ICDCN 2017**, Article 14.

**C21.** Poonam Goyal, Sonal Kumari, Sumit Sharma, **Dhruv Kumar**, Vivek Kishore, Sundar Balasubramaniam, and Navneet Goyal. 2016. A Fast, Scalable SLINK Algorithm for Commodity Cluster Computing Exploiting Spatial Locality. In **HPCC 2016**, 268–275.

**C22.** Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, Navneet Goyal, Saiyedul Islam, and Jagat Sesh Challa. 2015. Parallelizing OPTICS for Commodity Clusters. In **ICDCN 2015**, 33:1–33:10.

**C23.** Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2014. Parallelizing OPTICS for Multicore Systems. In **ACM COMPUTE 2014**, 17:1–17:6.

## Workshop Papers, Posters & Abstracts

**W1.** Stuti Sinha, Himanshu Kumar, Aryan Raju Mandapati, Rakshit Sakhuja, and **Dhruv Kumar**. 2026. The Language of Bargaining: Linguistic Effects in LLM Negotiations. In **ICLR 2026 Workshop on LLM Reasoning**. **arXiv (2026)**.

**W2.** Shubham Mishra, Shiv Tiwari, Samyek Jain, Gorang Mehrishi, **Dhruv Kumar**, Pratik Narang, and Harsh Sharma. 2026. From Facts to Conclusions: Integrating Deductive Reasoning in Retrieval-Augmented LLMs. In **ICLR 2026 Workshop on LLM Reasoning**. **arXiv (2025)**.

**W3.** Debdeep Sanyal, Aaryan Nagpal, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2025. time2time: Causal Intervention in Hidden States to Simulate Rare Events in Time Series Foundation Models. In **NeurIPS 2025 Workshop on Time Series Foundation Models (BERT4S)**.

**W4.** Utkarsh Tiwari, Aryan Seth, Adi Mukherjee, Kaavya Mer, Kavish, and **Dhruv Kumar**. 2025. DebateBench: A Challenging Long Context Reasoning Benchmark For Large Language Models. In **NAACL SRW 2025**.

**W5.** Shreya Bhatia, Tarushi Gandhi, **Dhruv Kumar**, and Pankaj Jalote. 2024. Unit Test Generation using Generative AI: A Comparative Performance Analysis of Autogeneration Tools. In **LLM4Code@ICSE 2024**, 54–61.

**W6.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin (Workshop Version). In **CCGridW 2023**, 308–310.

**W7.** **Dhruv Kumar**, Joel Wolfrath, Abhishek Chandra, and Ramesh K. Sitaraman. 2022. Towards WAN-Aware Join Sampling over Geo-Distributed Data. In **EdgeSys@EuroSys 2022**, 13–18.

**W8.** Yuanli Wang, Joel Wolfrath, Nikhil Sreekumar, **Dhruv Kumar**, and Abhishek Chandra. 2021. Accelerated Training via Device Similarity in Federated Learning. In **EdgeSys@EuroSys 2021**, 31–36.

**W9.** Yuanli Wang, **Dhruv Kumar**, and Abhishek Chandra. 2020. Exploiting Data Heterogeneity for Performance and Reliability in Federated Learning (Poster). In **SEC 2020**, 164–166.

**W10.** **Dhruv Kumar**, Aravind Alagiri Ramkumar, Rohit Sindhu, and Abhishek Chandra. 2019. DeCaf: Iterative Collaborative Processing over the Edge. In **HotEdge 2019**.

**W11.** **Dhruv Kumar**, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics (Abstract). In **SIGMETRICS 2019**, 15–16.

### Preprints (arXiv / SSRN)

**P1.** Rakshita Goel, S. Pranav Kumar, Anmol Agrawal, Divyan Poddar, Pratik Narang, and **Dhruv Kumar**. 2026. Domain-Partitioned Hybrid RAG for Legal Reasoning: Toward Modular and Explainable Legal AI for India. **arXiv (2026)**.

**P2.** Gaurish Trivedi, Alakh Sharma, Kartikey Singh Bhandari, Yash Sinha, Pratik Narang, **Dhruv Kumar**, and Jagat Sesh Challa. 2026. Trust Regions Sell, But Who's Buying? Overlap Geometry as an Alternative Trust Region for Policy Optimization. **arXiv (2026)**.

**P3.** Devanshu Sahoo, Manish Prasad, Vasudev Majhi, Arjun Neekhra, Yash Sinha, Murari Mandal, Vinay Chamola, and **Dhruv Kumar**. 2026. The Compliance Paradox: Semantic-Instruction Decoupling in Automated Academic Code Evaluation. **arXiv (2026)**.

**P4.** Aakash Trivedi, Aniket Upadhyay, Pratik Narang, **Dhruv Kumar**, and Praveen Kumar. 2026. A Hybrid Supervised-LLM Pipeline for Actionable Suggestion Mining in Unstructured Customer Reviews. **arXiv (2026)**.

**P5.** Aditya Thole, Anmol Agrawal, Arnav Ramamoorthy, and **Dhruv Kumar**. 2026. PhysicsSolutionAgent: Towards Multimodal Explanations for Numerical Physics Problem Solving. **arXiv (2026)**.

**P6.** Kartikey Singh Bhandari, Manav Ganesh, Yashwant Viswanathan, Archit Agrawal, **Dhruv Kumar**, and Pratik Narang. 2026. Actionable Advice from Reviews via Mixture of LoRA Experts: A Two-LLM Pipeline for Issue Extraction and Business Recommendations. **arXiv (2026)**.

**P7.** Kartikey Singh Bhandari, Tanish Jain, Archit Agrawal, **Dhruv Kumar**, Praveen Kumar, and Pratik Narang. 2026. A Multi-Agent System for Generating Actionable Business Advice. **arXiv (2026)**.

**P8.** Devesh Saraogi, Rohit Singhee, and **Dhruv Kumar**. 2026. Evaluating Novelty in AI-Generated Research Plans Using Multi-Workflow LLM Pipelines. **arXiv (2026)**.

**P9.** Aviral Gupta, Armaan Sethi, and **Dhruv Kumar**. 2026. TabPFN Through The Looking Glass: An Interpretability Study of TabPFN and Its Internal Representations. **arXiv (2026)**.

**P10.** Varun Bharti, Shashwat Jha, **Dhruv Kumar**, and Pankaj Jalote. 2025. Automated Type Annotation in Python Using Large Language Models. **arXiv:2508.00422 (2025)**.

**P11.** Gaurish Trivedi, Alakh Sharma, Kartikey Singh Bhandari, **Dhruv Kumar**, Pratik Narang, and Jagat Sesh Challa. 2025. HAEPO: History-Aggregated Exploratory Policy Optimization. **arXiv:2508.18884 (2025)**.

**P12.** Manan Gupta and **Dhruv Kumar**. 2025. Disentangling Polysemantic Neurons with a Null-Calibrated Polysemanticity Index and Causal Patch Interventions. **arXiv:2508.16950 (2025)**.

**P13.** Sankalp Tattwadarshi Swain, Anshika Krishnatray, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. Talking with Oompa Loompas: A Novel Framework for Evaluating Linguistic Acquisition of LLM Agents. **arXiv:2509.07389 (2025)**.

**P14.** Chinmay Pushkar, Sanchit Kabra, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. Beyond Single Bugs: Benchmarking LLMs for Multi-Vulnerability Detection. **arXiv (2025)**.

**P15.** Vaibhav Devraj, **Dhruv Kumar**, Jagat Sesh Challa, Parth Agarwal, Navya Kommuri, Trizal Garg, Prisha Singhal, and Dhruv Shah. 2025. CricBench: A Multilingual Benchmark for Evaluating LLMs in Cricket Analytics. **arXiv (2025)**.

**P16.** Unnati Saraswat, Tarun Rao, Namah Gupta, Shweta Swami, Shikhar Sharma, Prateek Narang, and **Dhruv Kumar**. 2025. Toward Intelligent Scene Augmentation for Context-Aware Object Placement and Sponsor-Logo Integration. **arXiv (2025)**.

**P17.** Arnav Gupta, Gurekas Singh Sahney, Hardik Rathi, Abhishek Chandwani, Ishaan Gupta, Pratik Narang, and **Dhruv Kumar**. 2025. Understanding Virality: A Rubric-Based Vision-Language Model Framework for Short-Form

Edutainment Evaluation. **arXiv (2025)**.

**P18.** Akshaj Prashanth Rao, Advait Singh, Saumya Kumaar Saksena, and **Dhruv Kumar**. 2025. Efficient Jailbreak Mitigation Using Semantic Linear Classification in a Multi-Staged Pipeline. **arXiv (2025)**.

**P19.** Aryan Chaudhary, Sanchit Goyal, Pratik Narang, and **Dhruv Kumar**. 2025. Investigating Spatial Attention Bias in Vision-Language Models. **arXiv (2025)**.

**P20.** Daksh Jain, Aarya Jain, Ashutosh Desai, Avyakt Verma, Ishan Bhanuka, Pratik Narang, and **Dhruv Kumar**. 2025. Large Language Models as Pokémon Battle Agents: Strategic Play and Content Generation. **arXiv (2025)**.

**P21.** Chayan Jain, Rishant Sharma, Archit Garg, Ishan Bhanuka, Pratik Narang, and **Dhruv Kumar**. 2025. Lights, Camera, Consistency: A Multistage Pipeline for Character-Stable AI Video Stories. **arXiv (2025)**.

**P22.** Darshil Chauhan, Adityasinh Solanki, Vansh Patel, Kanav Kapoor, Ritvik Jain, Aditya Bansal, **Dhruv Kumar**, and Prateek Narang. 2025. Bridging the Reality Gap: Efficient Adaptation of ASR Systems for Challenging Low-Resource Domains. **arXiv (2025)**.

**P23.** Charan Prakash Rathore, Saumi Ray, and **Dhruv Kumar**. 2025. Evaluating LLMs for Zeolite Synthesis Event Extraction (ZSEE): A Systematic Analysis of Prompting Strategies. **arXiv (2025)**.

**P24.** Arnav Ramamoorthy, Shrey Dhorajiya, Ojas Pungalia, Rashi Upadhyay, Abhishek Mishra, Abhiram H., Tejasvi Alladi, Sujan Yenuganti, and **Dhruv Kumar**. 2025. neuralFOMO: Can LLMs Handle Being Second Best? Measuring Envy-Like Preferences in Multi-Agent Settings. **arXiv (2025)**.

**P25.** Devanshu Sahoo, Manish Prasad, Vasudev Majhi, Jahnvi Singh, Vinay Chamola, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2025. When Reject Turns into Accept: Quantifying the Vulnerability of LLM-Based Scientific Reviewers to Indirect Prompt Injection. **arXiv (2025)**.

**P26.** Devanshu Sahoo, Vasudev Majhi, Arjun Neekhra, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2025. How to Trick Your AI TA: A Systematic Study of Academic Jailbreaking in LLM Code Evaluation. **arXiv (2025)**.

**P27.** Madhav Krishna, Dhruv Gupta, Kotra Keshav, Hritav Singh, Akash Bonagiri, Pratik Narang, and **Dhruv Kumar**. 2025. MarketMaven: An Agent-Based AI for Automated Startup Analysis and Market Intelligence. **SSRN (2025)**.

**P28.** Sasank Tumpati, Devansh Agarwal, and **Dhruv Kumar**. 2025. Magistrate: A Repository-Context Aware Multi-Agent Framework for Automated Code Review. **SSRN (2025)**.

**P29.** Debdeep Sanyal, Manya Pandey, **Dhruv Kumar**, Saurabh Deshpande, and Murari Mandal. 2025. Confidence is Not Competence: Evaluating Calibration in LLMs. **arXiv (2025)**.

**P30.** Debdeep Sanyal, Aakash Sen Sharma, **Dhruv Kumar**, Saurabh Deshpande, and Murari Mandal. 2025. Policy Optimization Prefers The Path of Least Resistance. **arXiv (2025)**.

**P31.** Henansh Tanwar, Kunal Shrivastva, Rahul Singh, and **Dhruv Kumar**. 2024. OpineBot: Class Feedback Reimagined Using a Conversational LLM. **arXiv:2401.15589 (2024)**.

**P32.** Aryaman Raina, Prateek Mishra, Harshit Goyal, and **Dhruv Kumar**. 2024. AI as a Medical Ally: Evaluating ChatGPT's Usage and Impact in Indian Healthcare. **arXiv:2401.15605 (2024)**.

**P33.** Vibhor Agarwal, Nakul Thureja, Madhav Krishan Garg, Sahiti Dharmavaram, Meghna, and **Dhruv Kumar**. 2024. "Which LLM should I use?": Evaluating LLMs for Tasks Performed by Undergraduate CS Students in India. **arXiv:2402.01687 (2024)**.

**P34.** Anishka, Diksha Sethi, Nipun Gupta, Shikhar Sharma, Srishti Jain, Ujjwal Singhal, and **Dhruv Kumar**. 2024. TAMIGO: Empowering Teaching Assistants using LLM-Assisted Viva and Code Assessment. **arXiv:2407.16805 (2024)**.

**P35.** Shreya Bhatia, Tarushi Gandhi, **Dhruv Kumar**, and Pankaj Jalote. 2024. System Test Case Design from Requirements Specifications: Insights and Challenges of Using ChatGPT. **arXiv:2412.03693 (2024)**.

**P36.** Ishika Joshi, Ritvik Budhiraja, Pranav Deepak Tanna, Lovenya Jain, Mihika Deshpande, Arjun Srivastava, Srinivas Rallapalli, Harshal D. Akolekar, Jagat Sesh Challa, and **Dhruv Kumar**. 2023. "With Great Power Comes Great Responsibility!": Student and Instructor Perspectives on the Influence of LLMs on Undergraduate Engineering Education. **arXiv:2309.10694 (2023)**.

**P37.** Atharva Mehta, Nipun Gupta, Aarav Balachandran, **Dhruv Kumar**, Pankaj Jalote, et al. 2023. Can ChatGPT Play the Role of a Teaching Assistant in an Introductory Programming Course? **arXiv (2023)**.

## Sponsored Research Projects

### 1. SPANDAN — AI/ML for Strengthening Police Investigation Processes

| | |
|---|---|
| **Funding Agency** | Gujarat Police (Office of Superintendent of Police, Ahmedabad Rural) |
| **Budget** | INR 34,80,000 |
| **Duration** | 11 months (03/2026–01/2027) |
| **Status** | Ongoing |
| **Role** | PI |

### 2. Revolutionizing Indian Higher Education Institutes with Generative AI

| | |
|---|---|
| **Funding Agency** | BITS Pilani |
| **Budget** | INR 20,00,000 |
| **Duration** | 24 months (11/2024–10/2026) |
| **Status** | Ongoing |
| **Role** | PI |

**Summary of Results & Publications:**

- Pathak, A.; **Kumar, D.**; et al. 2025. Rubric Is All You Need: LLM-based code evaluation. *ACM ICER 2025.*
- Arora, U.; **Kumar, D.**; et al. 2025. Analyzing LLM usage in an advanced computing class in India. *ACE '25*, 154–163.

### 3. AI for River–Climate Modeling

| | |
|---|---|
| **Funding Agency** | The Nature Conservancy (NGO) |
| **Budget** | INR 25,00,000 |
| **Duration** | 11 months (08/2024–06/2025) |
| **Status** | Completed |
| **Role** | Co-PI |

**Summary of Results & Publications:**

- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the role of land use–land cover for adaptive river basin management. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Targeting degraded hotspots in riparian corridors for rehabilitation. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the synergistic influence of LULC and climate on hydro-climatological dynamics. *Journal of Hydrology* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Disaggregating the influence of climate change and LULC on streamflow. *Journal of Environmental Management* (under submission).

### 4. Generative AI in Education Grant

| | |
|---|---|
| **Funding Agency** | WISE–IIE Higher Education Consortium |
| **Budget** | INR 30,00,000 |
| **Duration** | 10 months (08/2024–05/2025) |
| **Status** | Completed |
| **Role** | Co-PI |

**Summary of Results & Publications:**

- Akolekar, H.; **Kumar, D.**; et al. 2025. Generative AI tools in mechanical engineering education. *Scientific Reports* 15, 9214.
- Gupta, R.; **Kumar, D.**; et al. 2025. Sakshm AI: Socratic tutoring for coding education in India. *ACM JCSS 2026.*
- Goyal, H.; **Kumar, D.**; et al. 2025. Impact of LLMs on K-12 education in rural India. *Scientific Reports 2025.*
- Sanyal, D.; **Kumar, D.**; et al. 2025. Investigating pedagogical teacher and student LLM agents. *EMNLP 2025.*

## 5. Google Cloud Education Grant

| | |
|---|---|
| **Funding Agency** | Google Cloud |
| **Budget** | INR 12,50,000 |
| **Duration** | 12 months (01/2024–12/2024) |
| **Status** | Completed |
| **Role** | PI |